

Anurag Mishra

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PROFILE SUMMARY

- **Experience:** 7+ years architecting and deploying production-scale **Machine Learning, Deep Learning, Generative AI, and NLP** systems across retail, banking, telecommunications, and enterprise domains
- **Frameworks:** PyTorch, LangGraph, LangChain, CrewAI, AutoGen, LlamaIndex, FastAPI, Scikit-learn, NLTK, SpaCy, TensorFlow, Keras
- **Infrastructure:** Kubernetes, Docker, Git, PostgreSQL, MySQL, Redis, CI/CD Pipelines
- **Cloud Platforms:** Azure (OpenAI, ML Studio), AWS (Bedrock, SageMaker), GCP (Vertex AI), Linux
- **Certifications:** Azure Data Scientist, Databricks ML Professional, AWS Developer Associate
- **Technical Talks:** LLM Applications, Agentic Workflows, Multi-Modal LLMs
- **Publications:** Featured by LangChain, RAG Chunking Strategies, RAG Optimization, RAG Pipeline Architecture

EXPERIENCE

- **EY GDS** Hybrid
Manager - AI/ML Engineering Feb 2022 - Present
 - **Enterprise RAG System for 1500+ Bank Branches — 10K DAU:**
 - * Architected and deployed a distributed GenAI chatbot serving **9,000–10,000 DAU** for querying bank policies, circulars, and bulletins from SharePoint, reducing query resolution time from **5–10 min to <30s**.
 - * Engineered document ingestion pipeline using Azure Logic Apps with hybrid search (BM25 + semantic), cross-encoder re-ranking, and recursive chunking to process **6,000–7,000 complex PDFs** with tables and flowcharts, achieving **55% retrieval accuracy improvement**.
 - * Designed fault-tolerant backend architecture using FastAPI, LangChain, Azure API Gateway with **region-based load balancing, rate limiting, and graceful degradation**, reducing service downtime by **50%**.
 - **SLM Fine-Tuning for Automated Post-Call Actions — Contact Center Automation:**
 - * Fine-tuned **Small Language Models (Mistral 7B, Phi-3 etc)** using **LoRA/QLoRA** and PEFT on curated instruction datasets to generate structured post-call summaries aligned with enterprise SOPs.
 - * Deployed production inference on **vLLM** leveraging **PagedAttention, continuous batching, and prefix caching** for KV cache optimization, integrated with **ServiceNow APIs** for automated ticket generation and routing.
 - * Optimized throughput using **tensor parallelism and FP8 quantization**, reducing GPU memory footprint by **35%** and inference cost per call by **20%**.
 - * Scaled system to process **300–400 daily calls per center**, improving agent productivity by **60%** and reducing wrap-up time by **40%**, delivering **\$2M annual operational savings**.
 - **Multi-Agent LLM Orchestration for Customer Support Triage:**
 - * Architected stateful multi-agent system using **LangGraph** with **short-term memory (Redis checkpoint)** and **long-term memory (PostgresStore + pgvector)** for cross-session context retention and semantic retrieval.
 - * Designed specialized agents (**Intent Router, Information Synthesizer, Policy Validator, Escalation Negotiator**) with conditional routing and shared memory namespaces, reducing manual intervention by **60%** and improving triage accuracy by **40%**.

- * Integrated enterprise knowledge bases (SharePoint, Confluence) via **MCP with OAuth 2.1 authentication, RBAC-based scope controls, and token audience verification** for secure real-time tool orchestration.
- * Implemented evaluation framework using **Langfuse** tracking **Task Completion Rate, Tool Correctness, Agent Handoff Accuracy, and Semantic Coherence** for continuous agent refinement.

- **Deloitte (Office of US)** Remote
Consultant - AI/ML *Feb 2021 - Feb 2022*
 - **Transformer Fine-Tuning for Document Automation — 70% Cost Reduction:**
 - * Engineered transformer-based extraction pipelines using **LayoutLM and BERT variants** for automated document processing, reducing manual data entry time by **80% (200+ hours/month)**.
 - * Deployed models as **batch inference endpoints on Azure Databricks with MLflow model registry**, implementing CI/CD pipelines for model versioning and A/B testing, reducing operational costs by **70%**.
 - **Fixture Optimization for Retail Inventory — \$5M Cost Savings:** Developed mixed-integer linear programming (MILP) optimization solution to allocate optimal fixture counts per product and brand across retail chain stores, achieving **\$5M annual cost reduction**.
 - **Demand Forecasting System — 45% Accuracy Improvement:** Built hierarchical time-series forecasting pipeline using **XGBoost, LightGBM, and SARIMA ensembles** with feature engineering for seasonality and promotions, improving demand forecast accuracy by **45%**.
- **Virtusa Polaris** Hybrid
Machine Learning Engineer *June 2018 - Feb 2021*
 - **Trading Product Classification — 66x Throughput Improvement:** Developed multi-class classification system using **Gradient Boosting and ensemble methods** for automated trade activity tagging, scaling throughput from **15K to 1M classifications/hour** with **95% accuracy**.
 - **NLP-based PnL Commentary Validation — 80% Quality Improvement:** Built NLP pipeline using **SpaCy NER and rule-based matching** to detect inconsistencies in Product Controller commentaries, deployed as **RESTful API with async ETL processing**.
 - **Tech Stack:** Python, Scikit-learn, XGBoost, SpaCy, Pandas, NumPy, Flask, PostgreSQL

EDUCATION

- **National Institute of Technology, Manipur** India
Bachelor of Technology - Information Technology; GPA: 8.5/10 *July 2014 - June 2018*
Coursework: Data Structures & Algorithms, Machine Learning, Artificial Intelligence, Database Systems, Operating Systems, Computer Networks

HONORS AND AWARDS

- Excellence in Client Delivery Award, EY GDS (2023)
- Top 10 in Computer Science Department, NIT Manipur
- District Topper (Top 1%) in Higher Secondary Examination